

ActInf Textbook Group ~ Cohort 5 ~ Session 11

(Chapter 5, Part 2) ~ 11/28/2023

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all right welcome back everyone um it's our last discussion for this interval of activity with cohort 5 we'll discuss chapter 5 but also anything in the first half of the book just anything of what we've gotten up to so far then next week is our final discussion for the year we'll look at project ideas so here's where people have um written different different ideas that they they might want to work on and also we'll review future textbook groups everyone's welcome to fill out this Anonymous form and then also it's always cool to um see how many ideas people have had about the future textbook groups and what to tell other participants and also know that in 2024 when we return it's possible for people who want to explore having like a role in the textbook group we can explore that but that'll be for next week so for today um we'll go to chapter 5 or again any section of the book so does anyone want to just begin with any thought on five or anything that they've gotten up to in the first 100 Pages Susan go for it so I'd like um a little more elaboration on the high road and the low road I better understanding of how to differentiate those two yeah does anyone want to give a thought on that I um I still like one thing I said when we discussed this before um and it might be worth dialogue um the low road to me is all about what is the nature of inference um so how do we make sense of the world when the world is when we don't have I guess the base axioms um and then the high road is all about the nature of identity what does it mean to be an entity in a world and what's what are the first principles that constrain the capacity to be an embedded entity so it's really the the nature of the first question you ask and then um you have a bunch of follow on propositions that you get from that first question and both roads lead you to the same place which is active inference I like that so this so that's the um kind of what differentiates I guess um yeah perception and action so um so when you say the nature of identity um um is identity and you might be saying it on a couple of different levels there um you You' got the identity of who am I but also um you can expand that into groups and how they Define their boundaries am I tracking that the same way that you're thinking Andrew I would say so and I you can also take

an intrinsic view is just like when you get a sensory map how can you delineate it into separate things um and there's a process of identity going on there that allows you to label your sent field as a collection of separates so identity has like anytime that you can tokenize your senty field into something you've got a a an identity relationship going on can I ask a um what might sound like a um yeah um so where would you put degrees of freedom you where would you put degrees anytime you have multiple identities interrelate then you're going to have have a property of degrees of freedom of how they relate so I I guess you could pull it from either direction so it was almost in affordance yeah yeah okay and Daniel I'm just riffing here so yeah go for it okay yeah that's that's I have okay all all thanks for that I'll I'll give a little bit of a different um angle um the low road how you'd know you're on it is anytime some specific implementation is being described so that's like the actual constructive materialized basis of some inferential system it uses base theorem but JF clout here uses symbolic logic there's potentially other non- basing approaches people could try coming from the high road is the why you know you're on it because it's implementation Independence it's agnostic or factorized with respect to how you build it so these are talking about properties of different kinds of systems like system science or complexity science just general system properties that don't specifically say how it's built like I want the fast car that gets me from A to B I want a car made of aluminum but maybe the car made of aluminum is just a museum car and maybe the car that gets you from made of B is like made of paper um that's one answer how on the low road with the material substrate of the synthetic intelligence why on the high road um a little bit previous of a framework that was used before in the um like late 2010s when friston started using the high road low road um distinction there was uh an appeal actually to the more classical partitioning of cause with Aristotle oh funny typo with Aristotle's for cause material efficient formal and final so material cause is like the actual material like there's a house there because there's planks of wood the efficient cause the construction of the house the formal cause that which makes it a house and then the Tios or the final cause and this was used in the paper answering schinger question free energy formulation by ramstead at all 2018 and so here they explicitly took that kind of this is earlier on in their multiscale thinking and they explicitly applied this to tinbergens for Y which were like the biological adaptation of aristotles for wise so it it's especially when it's presented um this way in the textbook it's not for my understanding intended to be like necessary or sufficient or say that there's some one in only relationship between surprise minimization must be exactly right here between these two terms it's more just evocative principles and and hows that are actually used but if one

understands all the house if you understand all the building materials and all the properties then like the how and the why of active inference on this one road are fully defined so so that leads me to um so the in the context of I in fact if I I kind of wrote down the context of intent is actually the high road and the low road um would be you how I'm going to follow that the the scope of actually the um and I'm I'm toying or playing with or trying to understand discreet versus continuous and and I'm I'm I can see where they are um they are somewhat embedded in each other so it can't be delineated exactly but uh but in terms of the generative model modeling process is that what is that what this is pointing toward this is like a Prelude to choosing whether you are going to use a discrete or a continuous model this is discrete or continuous AG agnostic like you could you could go up and down yeah you could go up and down and you could have a fully discrete system or model and you could also have continuous time continuous steps yeah or you could have the hybrid model okay but it's an interesting question like is the continuous or discrete just a little difference or is it a big difference what this in so that's where I'm I'm trying to kind of put the scope of it um and and I may be ever complicating it but um but no thank you for that that's helpful yes thank you yeah especially as we come back to actual systems of Interest like in chapter five it's just interesting like okay this is kind of this is the road map in theory but where's any real system on this like do we drive to active then take take a turn and head over to the brain or is this our road map that we then transpose onto the brain I think there's a lot of space with like H how these more evocative schematics end up being used in a real setting can I take a I think it's more of a pedagogical skill tree rather than an implementation road map I agree with that I agree with that um though not explicitly linearly pedagogical but yes it's it's um like a mind map and it's a it's the map and not the terrain but at the same time it's not paved it's not bab it's a single lane dirt road Pony Express from f p down to B yeah and so when you when we use the term policy and I'm I may be conflating that with strategy and you know so and I'm I may be reifying the low road with strategy you say anything about that so policy is strategy contained in policy strategy let's even see if we use obviously it's not like a word has to be in this list to to be useful strategy can refer to a lot of different things policy most narrowly refers to the sequence of actions for a given time Horizon sequence of affordances that are planned to be taken over a given time Horizon so that would be down here with planning as inference so it's like the two fundamental features or or attributes that the thing has to have inbound dependencies which we interpret as perception outbound dependencies which we interpret as action now not all things plan like a pendulum is active but you don't need to appeal to planning whereas in the

case of things that do appear to either make sentient or Intelligent Decisions in the moment even if they're just one time step judgments or specifically engage with what appears to be planning like explicit considerations of multiple time steps and evaluation against Alternatives and abductive reasoning and all of this then how is active inference going to how hash low road How is active inference going to approach this question of planning and action selection as an inference problem in fact part of a unified inference problem with perception so action selection and planning and policy selection is down here in the basian branch and then the way that we would kind of touch to policy or say say what what is the strategy by which the actions are selected coming from the high road without talking about how we're going to do that inference it would be like well if something is going to be self-organizing itself to persistently minimize its surprise amidst environmental regularities irregularities maintaining its Markov blanket and self- evidencing its own existence on its own body and Beyond through Niche construction then it might have to do some kind of planning where it might have to is like saying like you know it'd be nice if somebody made this dish but it's high road because you're not saying how that's going to happen um any anyone who hasn't raised anything like want to type or ask a question here's a nice part of five that that they're discussing it in the neuromodulation case however this applies outside the brain like why why would we even make maps why would we associate features of our Maps generative models with anything that is actually there in the territory like why do it it lets us explain observe biology form hypotheses and develop non-invasive methods also not um not only or exhaustive like another kind of common refrain would be like explain predict design control or appreciate influence control developing explanatory accounts that is about kind of getting our desk organized organizing the data we already have what we already know developing a more coherent or more concise account forming hypotheses and and developing methods are moving more into the forward-looking space you could do that without making an account first but usually catching up with the past then helps develop the future goinging components of the scientific aspect like a given account of the brain here's what the brain does and then here's the methods that people bring to bear Susan I really like how you Associated um the biology with self-organization um observed biology so that brings up for me the observed biology um is it we're not limiting this to what's what's observable or are we for the mapping for the generative process mapping I mean I you it was helpful for me quite frankly to listen to the session you just posted on chapter six yesterday because it kind of brought home what is this for and so that generated a lot of new questions so you know my question when you say obs OB served

biology um and you forming hypothesis is a little bit uh more tangible but observed biology that that term yeah it's like okay that's where the self-organizing is yeah is the structure it's a funny way to say it like um that it's observed I wonder if it's almost like a little um like let let's think about the fundamental challenge of the a matrix to take observations and map them to Hidden States so now thinking about the scientific Endeavor or scientist or just sense maker in terms of active inference there's observations and and then there's the unobserved hidden States there's like the estimates of what those observations reflect thermometers in the room measurements and then the mapping to the real temperature so talking about let us explain observe biology observed connects us to empirical observations like well the person who was coughing the PCR test had this result the person who wasn't coughing the PCR test had that result that's an observation but then the explanatory account obviously includes things that aren't simply observed otherwise the quote data would speak for themself but they never do that's why there's like science and law and all of this so This points both ways because it's saying we are trying to develop an account of observations now those might be super super sparse observations we have one video of a Jaguar one time doing this move so it's not that observation means we need to wait until everything is fully observed but there's something that explanation points to that isn't just like an inventory of what data were collected otherwise would say it lets us create an inventory of what we have measured but an inventory isn't an explanation so the form hypothesis is actually in the interpretation of perception absolutely absolutely helmholtz perception as unconscious inference it's like there's there's a blur a red blur and there's uncertainty but it's like well if somebody is at the racetrack maybe it's a red car maybe they have a strong prior but that's a hypothesis but maybe it turns out it was actually just you know an interior designer moving a Swatch of color in front of their eye so then then later on they they they up evaluate the probability of that hypothesis how are those hypotheses generated and winnowed abductive inference so whether we choose to kind of um exalt it and make it clean and cut and dry like a scientific hypothesis or whether to blur it back into the kind of imperceptible like could a hypothesis does a hypothesis need to be experienced um if if it doesn't and one's willing to kind of relax that and use hypothesis to include just um proposed causes proposed latent causes of observations then that is exactly what these accounts are I mean what's what does the brain do propose and select amongst hypotheses again in this more broader sense not just the scientist sense of observations I'm my my stomach is rumbling and this and that and this and that maybe I'm in this physiological state Sharon oh go ahead so one final question and uh and it may come out in somebody else's

question but um but hidden States would would invariably based on my understanding at this point in time um lead to probability so because if it's a hidden State we can't know it and and um and so there we have to kind of rely on what's the probability of this hidden State impacting the um the hypothesis would that be correct yes okay like just just to to um to make hyperbole usually the data are taken without error and the hidden States always come along with a probability distribution this is just the simplest case like thermometer says like 72 you might know that it's like the most joke inaccurate thermometer ever but you still take that 72 and you write it down your lab notebook as 72 and then you have some probability that's like okay well my estimate of the temperature is actually 80 plus or minus 10 because I know this thermometer is noisy and it low balls so you have some probability distribution for the hidden state but then usually the data are taken as simply the value that they represent and that's reflected with the Dirac Delta distribution that you'll hear people talking about the Delta distribution and that's basically a special statistical distribution that's like flat everywhere and then just has a spike at the one Loc so that's kind of a way to coerce like a data point into a distributional framework that came up a lot in the message passing live streams in 55 with Magnus um sharan or anyone else want to like give a thought yeah hypothesis is used a lot of times ranging [Music] from like the view from the outside like talking about the basing brain hypothesis as a scientific idea to this more like view from the inside like the agent is hypothesizing whether it is an apple or a frog in their hand here uncertainty about the hidden state is being explicitly described as a range of so would the the Delta distribution be a way to account for that range of hypothesis uh is that what is that what you're saying all go you get into so much probability that you know then then just like overwhelmed with probabilities well yeah that can happen let's look to um I'm going to the slides for for 55 because this is where like there's kind of an interchange between the uh okay check out the the papers or or 55 series for for more information but basically what is happening on this is a a factor graph so it it has a a one-to-one relationship with a Bayesian graphs that we look at in the textbook um at time point one just looking only at this one node that's been kind of highlighted um here there's an edge and here there's a little Delta land like a little token Delta that lands on that edge so before um the data have been collected at all this is like before the first data point this is after the first data point and this is after the nth data point or the teth data point so before this is kind of like an this is like a a bungee chord that's not pinned down there's just a full distribution that relates these two components then the data point comes in and then that is seen as a clamping so it's kind of like the bungee cords waving around the probability

distribution could be within this range and then the data point lands like the dart lands and it just pins the cord right there and then um to kind of keep the kind of that thing the same the Delta distribution is used so it's just kind of like saying we're going to clamp whatever distribution we thought was going to be there um instead of it being a gaussian with 10 plus or minus one then the data came in it was 11 now we have a derck distribution with a parameter of 11 so it kind of keeps that distributional flavor and Keeps Us in this probabilistic setting and what's Happening Here is basically before any data come in the system is totally unclamped and then as data points come in those they clamp specific parts of graph and then inference proceeds on the unclamped Parts I don't know if it'd be helpful but but think about like a guitar like when you clamp down on a string and then the rest of the string vibrates in a different way so an action has been taken or a decision's been made I mean would be one segment one example not necessarily the only ones but in other words when she's clamp down there's movement yeah yeah actions have been taken time by time step by time step here specifically what's being highlighted is like the sensory um data point coming in but at each time step it's not just that a sensory data point comes in also that the action is taken becomes irreversible to a certain degree yeah like it um the past you becomes sedimented and reflects in the current state of the present because then you could say well what if you forgot the past data point well that's kind of like unclamping the past and then now you have another uncertainty distribution over memory like yeah what was the temperature last week I threw out the data or could that represent you're falling into a pattern or a habit that's that's an an interesting question also probably many ways to answer it but like one way to look at it from um from the kinds of models we looked at in the textbook would be like um if we were going to if we were going to if we were gonna have a a memory of um if we were gonna have a memory of um we're we're P pulling B balls out of an urn this is a very common statistical setting like you're pulling a ball out of an ear um and one one way to um you know it's coming up you know orange orange green orange orange green like two-thirds of them are orange or something so one kind of memory that you might want to have would be like remember which time step you pulled which ball and then your memory register is going to grow linearly accordingly to how many polls that you make that you're going to keep on recording the full data so that's good for for to to that that is the full data that's going to enable the fullest analysis but let's just say that you in a um computation limited setting or maybe you wanted to respect the privacy of previous polls then maybe all that you would carry forward would be like um 2third of all the polls that I've made are orange and I've made a 100 poles so then you've coar grained over like which

specific polls were which but you have now all you carry forward is these two numbers how many polls have you made and what fraction of them were Orange um so now you have a smaller memory register and that is exactly how one way that people have modeled how habit is learned so instead of the agent needing to memorize the exact total sequence of all previous actions and then do this reanalysis in the moment with like the whole sequence of action um you can sediment previous actions into habit and then you just carry the Habit e forward as your policy prior so that is one lower computational requirement way to carry forward and slowly learn habit thank you yeah that's interesting though and then that statistical setting is called the POA urn there's so many variants of this statistic problem but as with many statistical problems like we have observations that's the color of the ball and then we have our hidden State we can't see inside the urn all we can do is draw and so in this one um it's not emphasizing the policy selection planning side like the the the draw just is given to you but um we could look at the um the related setting of a bandit or a multi-armed bandit which is kind of like a semi joking term for a slot machine I guess because it steals your money one-armed Bandit but the multi-armed Bandit is like a bunch of slot machines and you don't know what's inside of them slot machines like the urn um now Which slot machines should you use one strategy is just sit down at the first one keep using it another strategy is just rotate through all of them equally another strategy is um sit at them proportionally to how rewarding they are are another one is sit at them proportionally to how confident you are about your estimates in them or inversely confident to how um inversely timed to how confident you are in them and so that slightly more active setting of choosing where to draw observations from that is where we've seen some very useful active for um anyone who hasn't asked a question W to it could be about any aspect of the first half of the book or the whole book I have some project Associated questions but I people with related questions first yeah let's look look at the project here or where well first um kind of off of a question you asked during live stream 55 um where you were asking about active inference toolboxes um do you have any thoughts or recommendations if folks want to start playing around in a computational setting on active inference principles of what toolbox we should start with or yeah good good so few few answers first just to summarize in 55.1 or two or whatever it was like I asked Magnus kud all about the toolbox and about like the ution of the toolbox and he gave a truly classical highlights answer which is like in terms of software packages from the 90s to the early 2000s it was all about mat lab language statistical parametric mapping toolkit with a special Focus from neuroimaging created and maintained by Carl friston's neuroimaging group at UCL then

then over the last 5 years as more machine learning connected with active inference Connor Hines Alex shance stafne dcus and others worked on pmdp in Python bringing many routines from SPM into python setting and then most recently with Bert of rise's group with Magnus at all they are developing the Julia toolkit which is even more General than active inference you can do bayyan settings that aren't active inference um or are just super derived special cases in terms of um where to start or look in the code page um here there's like 30 or 40 different code implementations for ACM and related I always add whenever I find them and anyone else can add them to um if you are fluent in the language then it may be enough otherwise even if you're not familiar with a language copying it into a language model and asking for explanations might be cool or asking to translate it into pseudo code or give an example all these other kinds of fun things so these are um some specific places to start that's com kind of jumping in in the computational end which is going to be perfect for some I think the other toolbox that's now provided is chapter six and that's more like the um the um Cafe sketchpad toolbox like when you're sitting down to design an active inference model and it's not just just well which computational architecture should we use but maybe we want to walk up to that point or realistically develop an artifact if we address these questions I think we're already in a situation where this provides enough information to a language model to develop the computational generative model so we have the classical tool box elements and if somebody can just if somebody's already in the mechanic shop and can just get their hands dirty on these tools they'll find more than enough and for those for whom just jumping into this kind of scripting isn't intuitive I think now we have a more informal style that's almost conversational that can get us there too especially on teams does that address it or what else would be helpful that was helpful um follow on question do you know if versus um open sources or publishes any of their stuff or I assume that's pretty proprietary them offand I have not like seen any specifically released code but it it may be but I'm I'm not sure yeah it be added here as soon as I find um in machine learning Street Talk mentioned some something where they have like an algorithm that can kind of identify agents which is or entities um that was very interesting to me and something I was hoping to try to make a um implementation of so that's why I asked yes interesting I remember that too Susan I'm in in their um in versus uh video which was kind of an introduction they distinguish between open Ai and um and their program as being open source so one thing you might want to do Andrew is sign up for their beta which is on their website already done yeah and so so we'll see how what is open source and what's not yeah it is an an interesting question what is open source like just just not even with ACM with

language models like for some people it's like you can use it it's open source what's the issue whereas for other people it's like no the Full Source has to be provided the full data set that was used for training and the parameters that were used for training and the weights of the model have to be provided like Linux kernel the source code is there and you can compile it yourself that's what makes it open source and then even further back outside of the artifact of Interest like is the mailing list Open Source can people be observing or even included in those discussions like those are all I think very interesting um I'll add that um it if um to understand how red hat makes money might uh might inform you because you what red hat U banks on is that what is open source is not necessarily reliable and so they've actually taken the open source and turned it into something that people can can depend on so I don't if that's helpful or not read where versus is going but I mean to understand that model is helpful yeah yeah definitely um interesting and even even though we expect and prefer the open source materials to be reliable and anybody who wants to for example have access to the um textbook group repo at The Institute GitHub that's one great place if people just want to play around with code and don't just want it to be like a little abandoned repo on their own but maybe want to go in you know chapter by chapter and cohort after cohort will iterate towards having reliable pieces like there will still be more and it's not that it's our only um Guiding Light but having reliable open-source um accessories to the textbook and to other educational materials will be really important um anyone who hasn't raised anything wna uh probably most of the ones with get uh will be open source um I'm just looking at uh RX infa and that's got a um MIT license oh yeah so uh the moment you get on to GitHub do you find that as good or bad just out of curiosity uh what the fact that it's it or the or MIT I've Okay I uh my technical background as a commercial coder I've never really worried about the different subtleties of MIT versus uh something that's um uh approved um is MIT less restrictive about copying you can make closed Source copies I suspect um Okay so is not so maybe not so good at supporting the open C open source cuse but in terms of just being able to use it I think that would be uh I wouldn't have any worries about that on my somewhat limited um thank you experience yeah interesting I also don't know there's I know there's many flavors of Licensing um yeah this repo and their documentation website is is is extremely wellb built second in being well-built would be pmdp resources and documentation and then kind of in a class of its own would be the SPM documentation which has textbook and the manuals and a list serve however those discourses tend to be related um human neuroimaging experiments so you have less of like people asking about like how how do you do this like theoretical agent

identification or something like that or some extension and more of just like I'm using this machine with this experimental setup how do I get this to work but yeah it's amazing how many are out here and so anyone who finds or makes more please please add them yeah pmdp has a BSD three Clause license which again is um uh very open source uh so yeah here's the PDP documentation and um follow the link to GitHub you'll find uh the license cool yeah also this this documentation this collab if people just want to run they they don't even have a comp a computer with a coding setup you can run it in your browser and these are really useful um Andrew yeah I have a um unrelated question that's um for you specifically Daniel um related to a paper you wrote in 2020 called active inference and behavior engineering for teams yep um has there been any Bon work this is also very related to my interests um so I was just wondering if you have a uh any you can mention yes the follow on work was the active inference Institute oh cool all right indeed then known active inference lab basically just to kind of close it this was September 2020 and then we thought okay where now biggest issue was the accessibility rigor and applicability of the framework because getting to this artifact was not so hard but to move from multi-agent model model of of Team narrative communication cyber physical systems into having them we were like whoa how far even is that hence the lab okay thank you all it's been a great first section will come back at the later time next week and then we haven't set the dates or the times for second half of cohort 5 or for anything else if people want to like explore any role for a 2024 textbook group we can talk about that so just email if anything comes up otherwise thank you and peace bye and